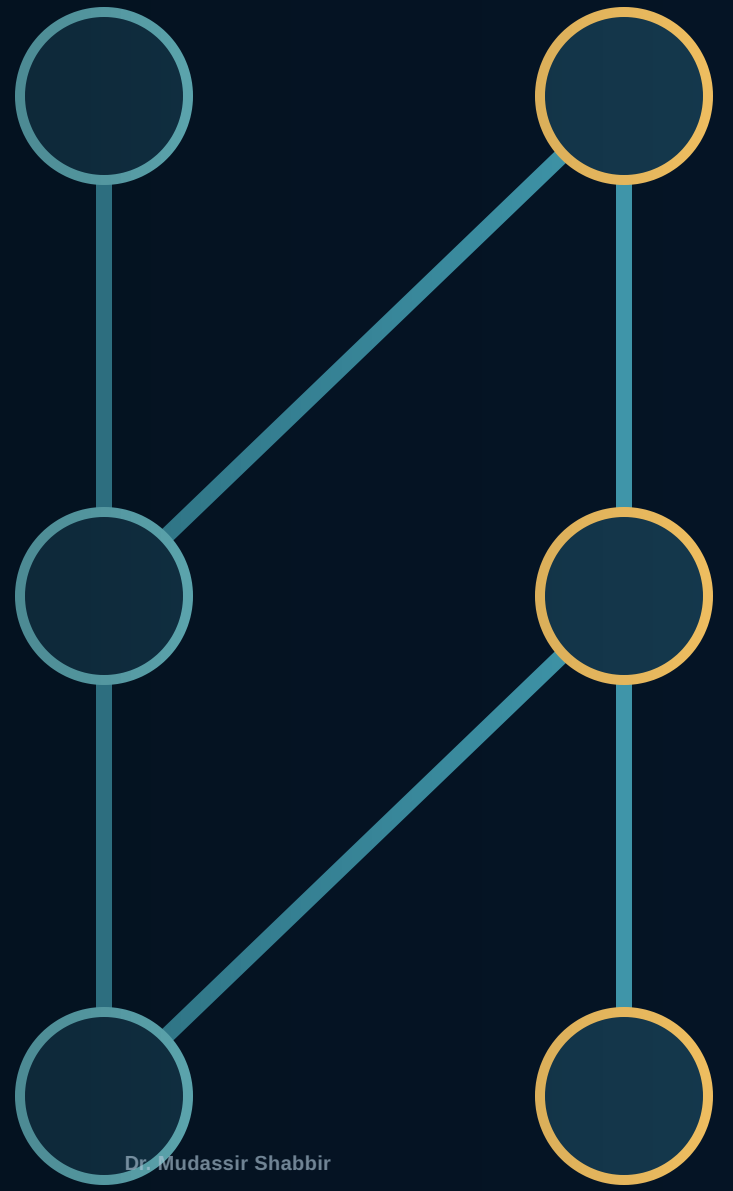
02 · HASSE DIAGRAMS

Draw only the essential edges

A Hasse diagram stores the comparisons that force all the others.



A cover has no intermediate element

Write $x \prec y$ for $x \preceq y$ and $x \neq y$.

Keep the cover

y covers x if $x \prec y$ and no z satisfies $x \prec z \prec y$.

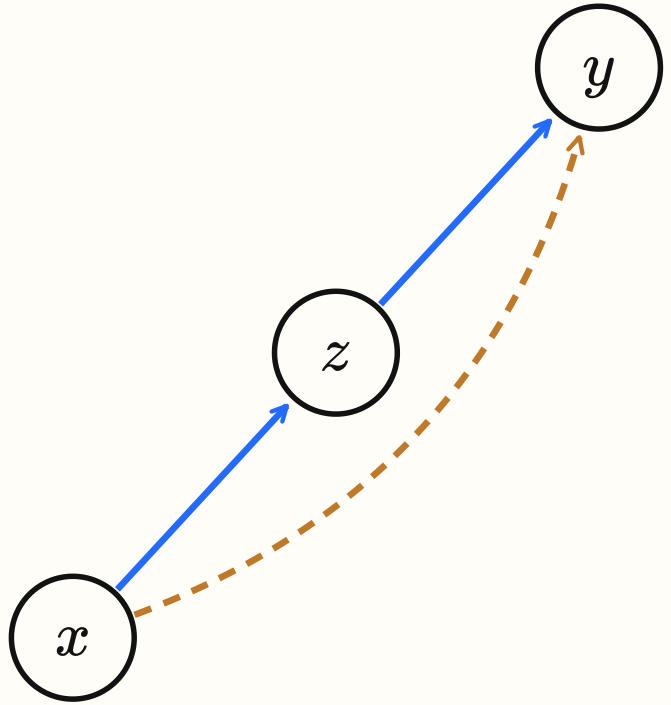
Remove only what is implied

Omit loops and transitive shortcuts; put the greater endpoint above.

Read upward

$x \preceq y$ means $x = y$ or an upward path leads from x to y.

An intermediate element rules out a cover



$x \prec z \prec y$: y does not cover x. The solid edges are covers.

For finite posets, covers determine the entire relation.

Constructing a Hasse diagram

$S = \{1, 2, 3, 4, 6, 8, 12\}$, ordered by divisibility. Begin with $a \to b$ whenever $a \mid b$.

00 Arrange vertices by level

Place each greater endpoint above its smaller endpoint.

01 Remove the loops

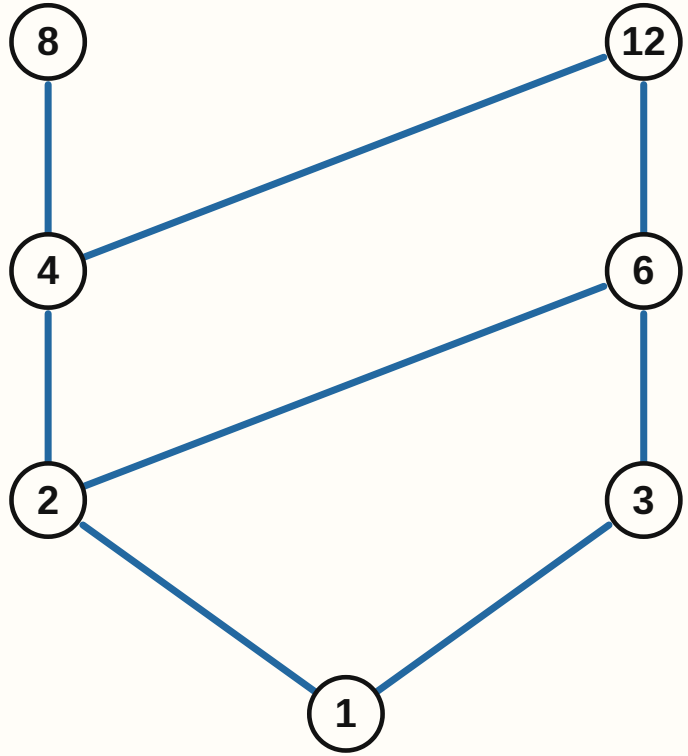
Every element still divides itself.

02 Remove transitive edges

An upward path already implies the comparison.

03 Keep covers; omit arrowheads

Keep eight covers; read every edge upward.



Hasse diagram · 8 cover edges

The relation stays the same: $a \mid b$ means $a = b$ or an upward path from a to b.

EXERCISE 1

An upward path is a comparison

Use the Hasse diagram of $(\{1, 2, 3, 4, 6, 8, 12\}, |)$.

01 **Is $2 \preceq 12$ even without an edge?**

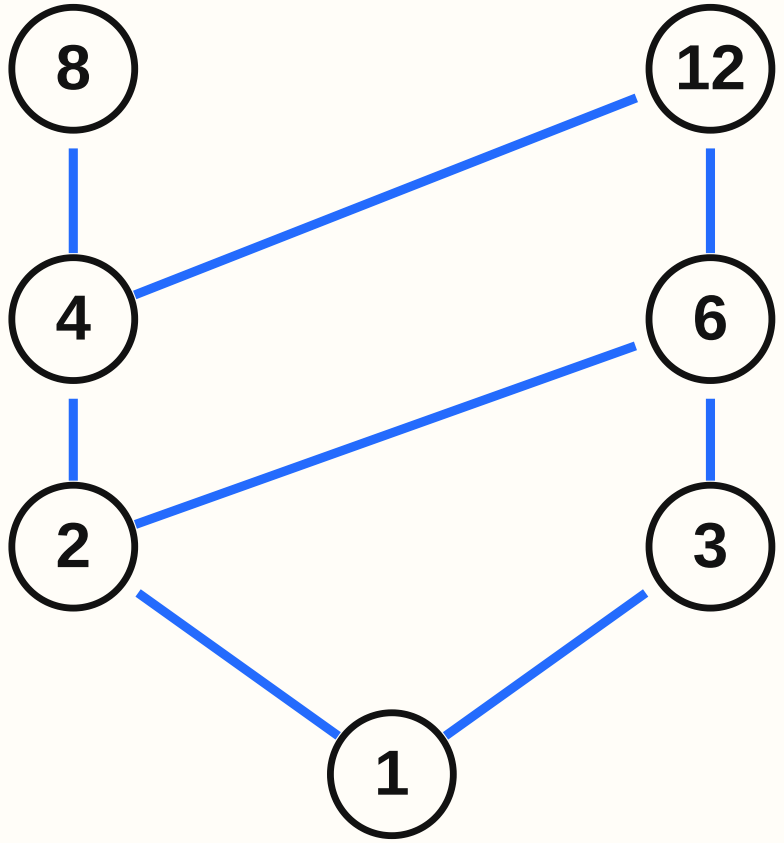
Yes: $2 \to 4 \to 12$ (also $2 \to 6 \to 12$).

02 **Are 3 and 4 comparable?**

No upward path connects them in either direction.

03 **Does being drawn higher make 8 greater than 6?**

No. Height alone is insufficient: $6 \nmid 8$ and $8 \nmid 6$.



EXERCISE 2

Build a Hasse diagram from scratch

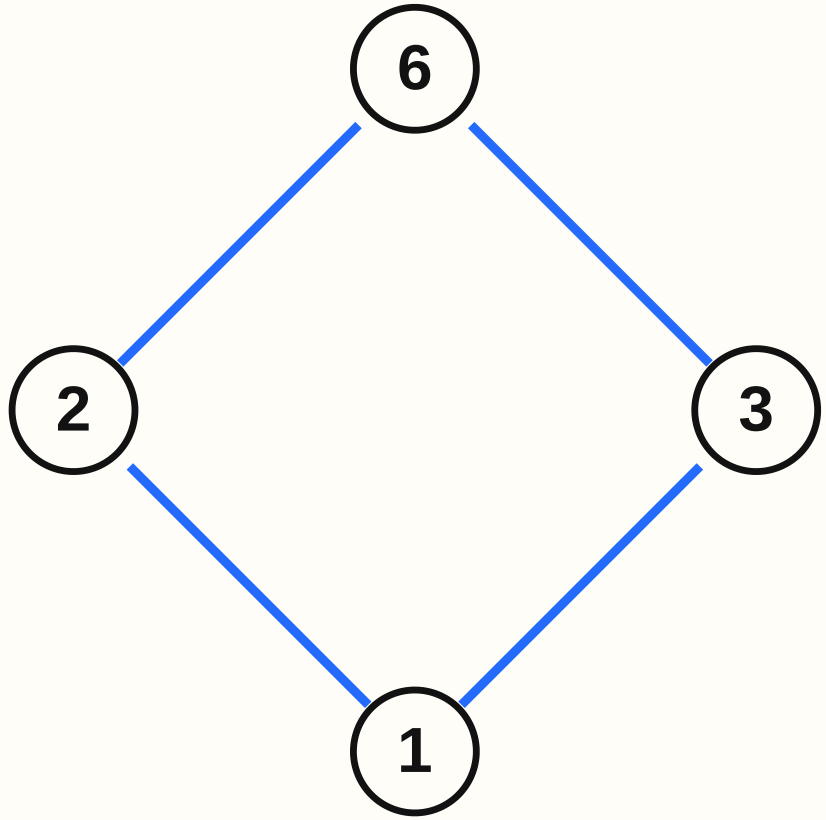
For $S = \{1, 2, 3, 6\}$ ordered by divisibility:

01 **Which pairs are covers?**

$(1, 2), (1, 3), (2, 6), (3, 6)$.

02 **Should you draw an edge from 1 to 6?**

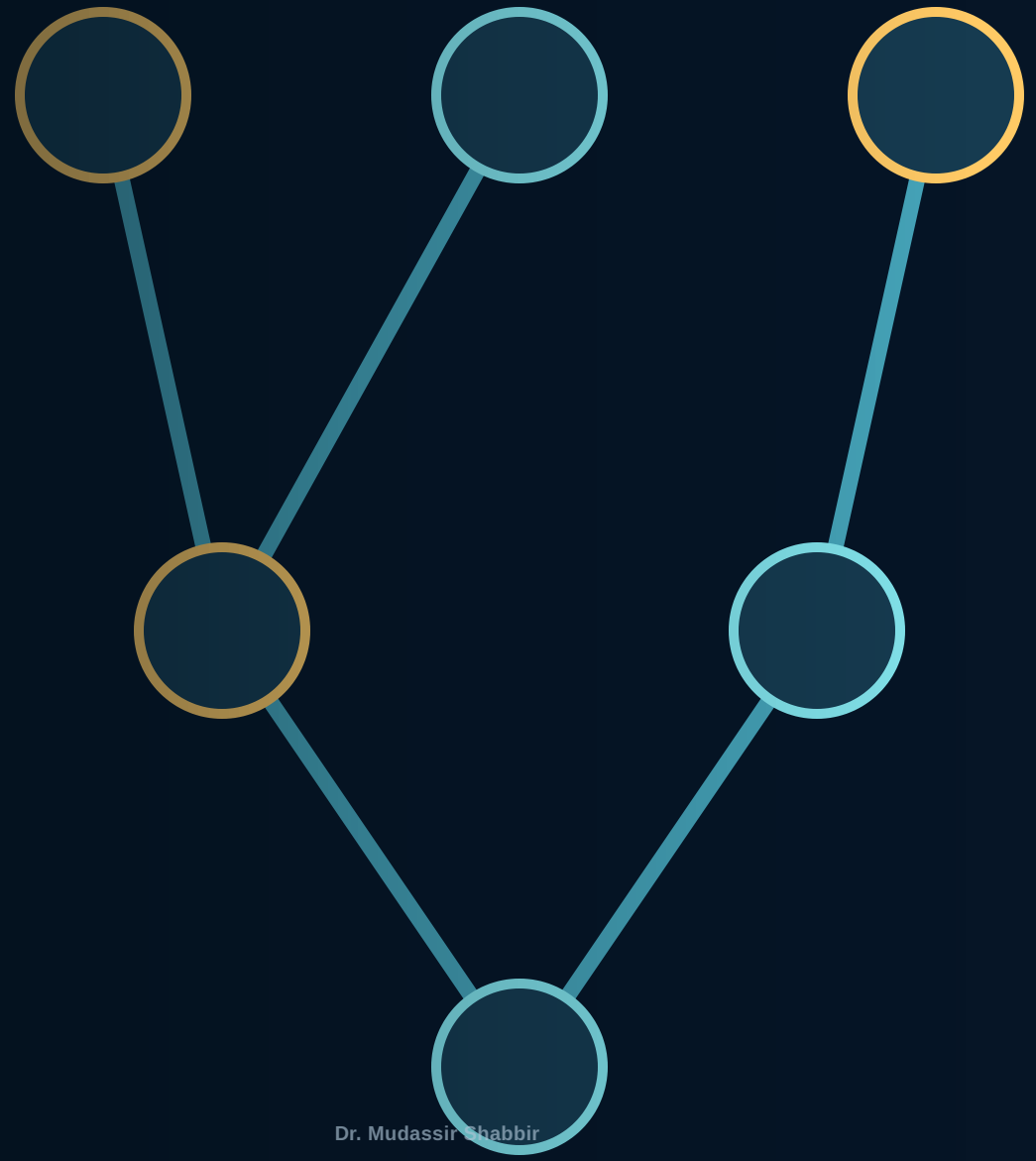
No: there is an intermediate element, 2 (or 3).



03 · EXTREMA

At the top of which order?

An element can be maximal without being greatest.



Minimal and maximal are local conditions

Minimal

No $y \neq x$ satisfies $y \preceq x$.

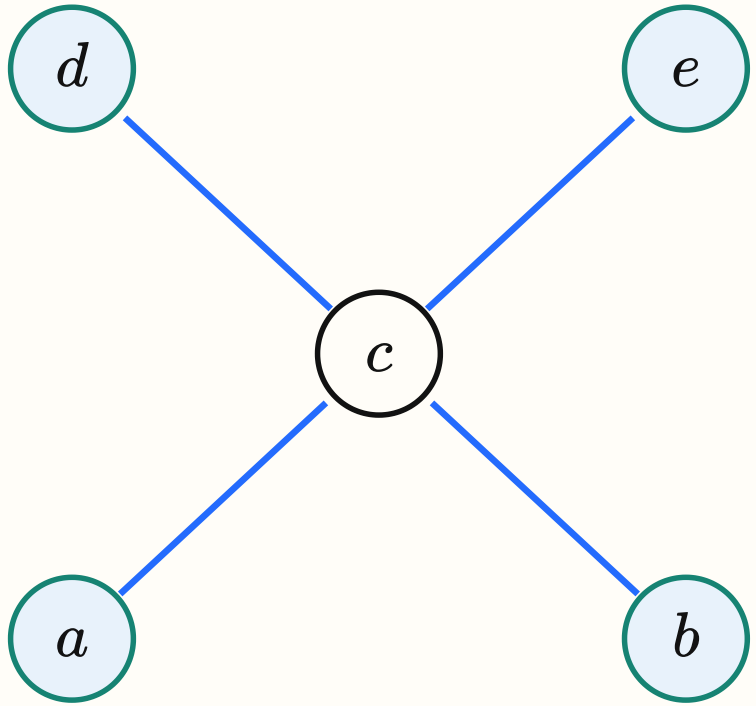
No element sits strictly below x in the order.

Maximal

No $y \neq x$ satisfies $x \preceq y$.

No element sits strictly above x in the order.

Several extrema can coexist



Minimal: a, b. Maximal: d, e. Neither a least nor a greatest element.

A finite diagram: minimal vertices have no edge down; maximal vertices have no edge up.

EXERCISE 3

Find every minimal and maximal element

Work in $(\{1, 2, 3, 4, 6, 8, 12\}, |)$.

01 **Which elements are minimal?**

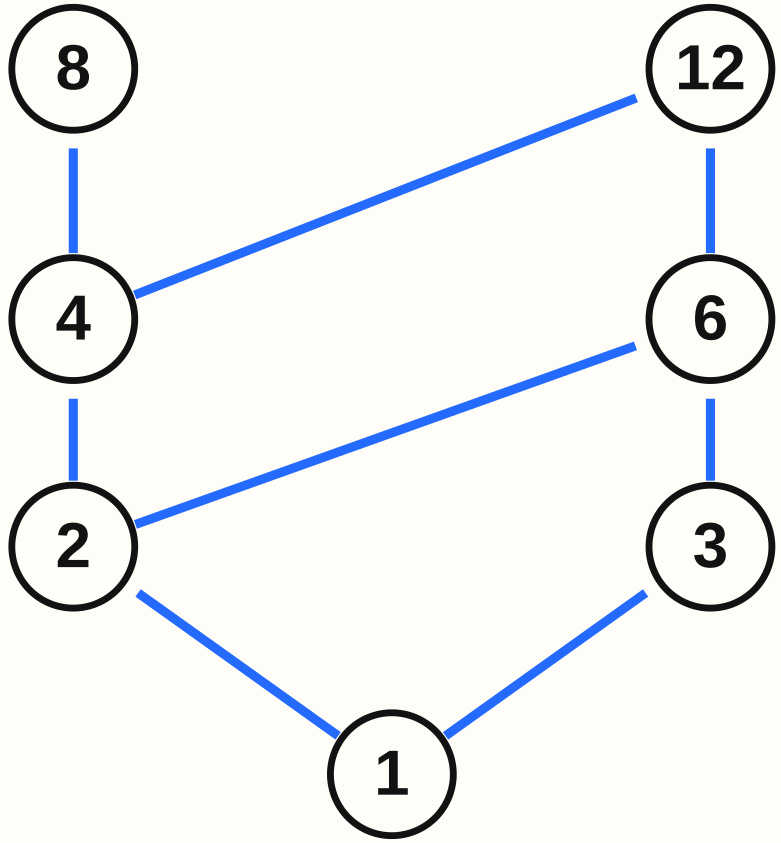
Only 1: every other vertex has something below it.

02 **Which elements are maximal?**

8 and 12: neither has an upward edge.

03 **Why is 8 still maximal when $8 < 12$?**

Maximality uses divisibility, and $8 \nmid 12$.



Least and greatest compare with everyone

Least element

$\ell \preceq y$ **for every** $y \in S$**.**

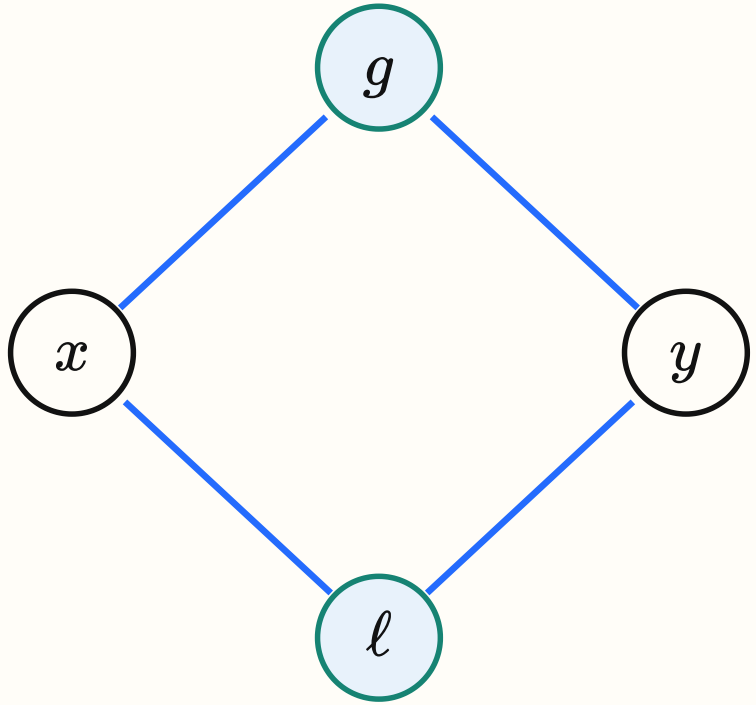
A least element lies below every element.

Greatest element

$y \preceq g$ **for every** $y \in S$**.**

A greatest element lies above every element.

Reach every vertex by an upward path



ℓ is below everyone; g is above everyone.

Least implies minimal. Greatest implies maximal. The converses fail in general.

EXERCISE 4

Two maximal elements; no greatest element

Use the same divisibility poset.

01 **Does a least element exist?**

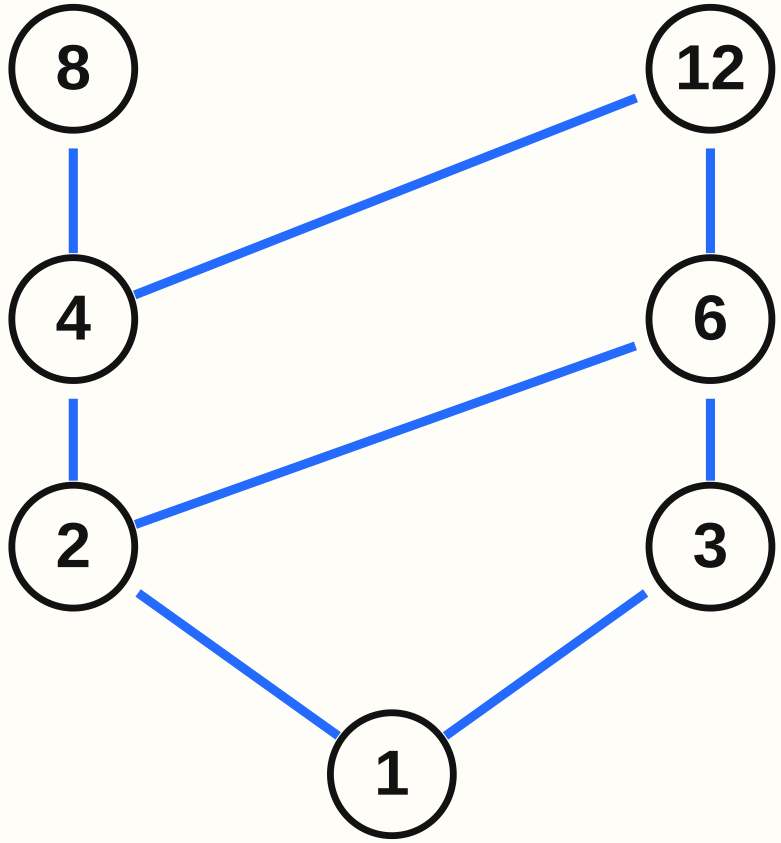
Yes: 1 divides every element of S.

02 **Does a greatest element exist?**

No: neither 8 nor 12 is divisible by the other.

03 **Could 24 be the greatest element here?**

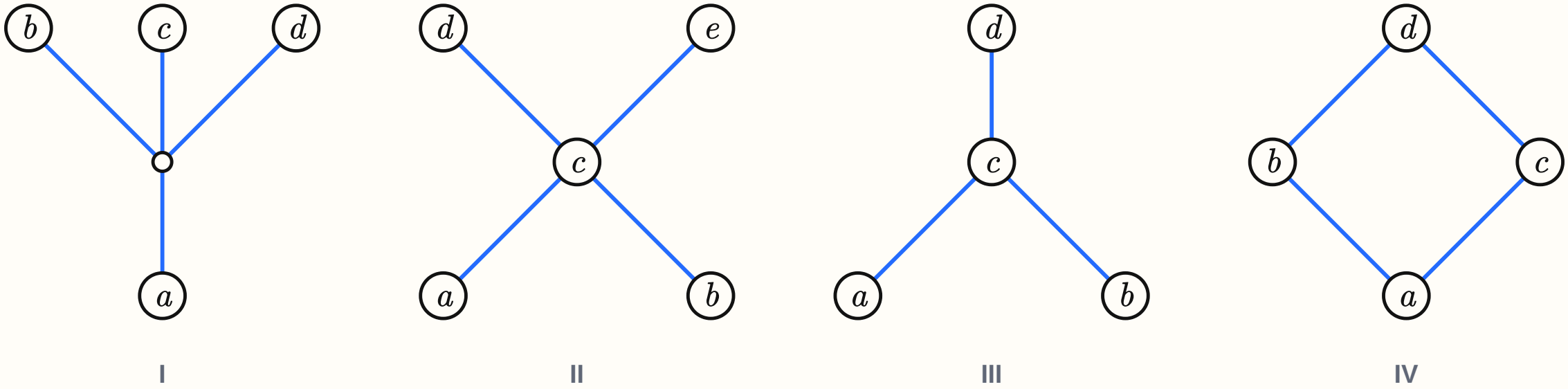
No: $24 \notin S$. Extrema must belong to the poset.



The numerically largest member need not be greatest in this order.

EXERCISE 5

Read four different outcomes



I II III IV

01 **Which diagrams have a greatest element?**

None; none; d; d.

02 **Which diagrams have a least element?**

a; none; none; a.

A greatest element cannot have a rival

Suppose g and h are both greatest in the same poset.

Use the definition twice

$g \preceq h$ because h is greatest; $h \preceq g$ because g is greatest.

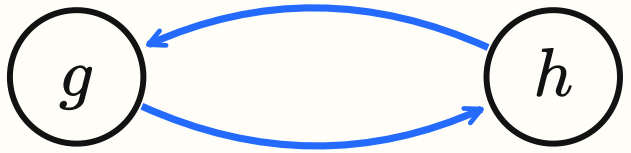
Use antisymmetry

$g = h$.

Reverse the argument

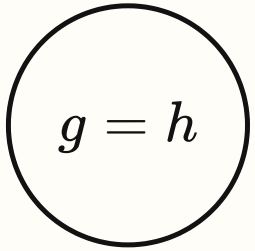
A least element is also unique, when it exists.

Suppose both are greatest



$g \preceq h$ and $h \preceq g$.

Antisymmetry identifies them



They are the same element, not two rivals.

Several maximal elements are possible; several greatest elements are not.

The underlying set changes the answer

All subsets, ordered by containment

$(\mathcal{P}(U), \subseteq)$

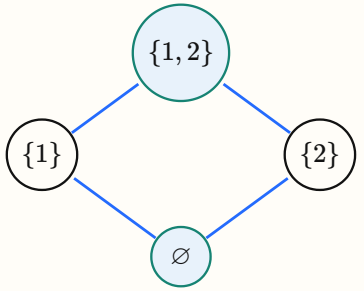
Least: $\varnothing$. Greatest: U.

Positive integers, ordered by divisibility

$(\mathbb{Z}^+, |)$

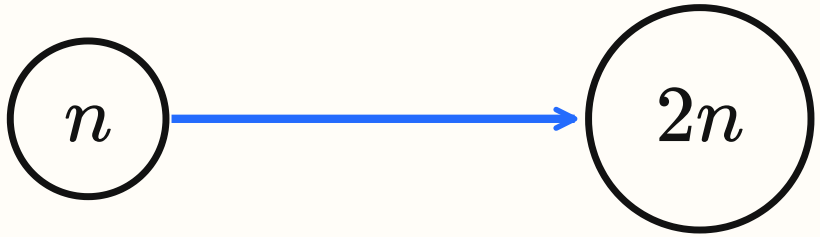
Least: 1. No greatest: $n \mid 2n$ and $n \neq 2n$.

Example: $U = \{1, 2\}$



$\varnothing$ is least; U is greatest.

Every positive integer has a successor



$n \mid 2n$ and $n \neq 2n$; no greatest element.

Always specify both the set and the relation.

EXERCISE 6

Infinite orders may have no extrema

Compare these familiar posets.

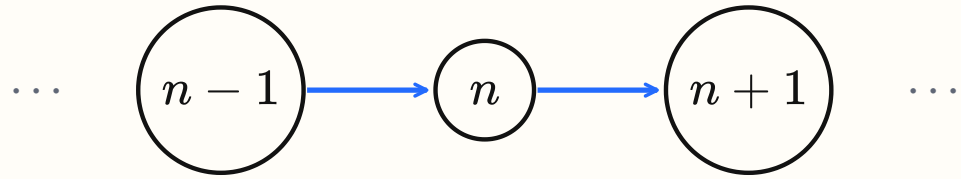
01 **In $(\mathbb{Z}, \leq)$, find all minimal and maximal elements.**

None: for each n, $n-1 < n < n+1$.

02 **In $(\mathbb{Z}^+, |)$, find all maximal elements.**

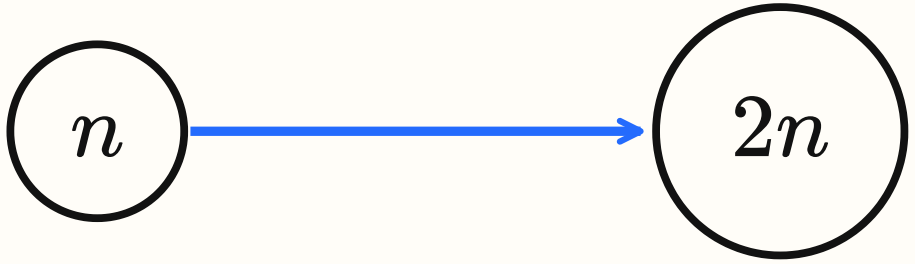
None: every n has the strictly greater element $2n$.

$(\mathbb{Z}, \leq)$ continues both ways



For every n, there is a smaller and a larger integer.

$(\mathbb{Z}^+, |)$ always extends



Every n has a strictly greater element $2n$.

EXERCISE 7 · OPTIONAL CHALLENGE

How large can an intersecting family be?

Let $n \geq 1$. Distinct subsets $A_1, \ldots, A_k \subseteq U = \{1, \ldots, n\}$ satisfy $A_i \cap A_j \neq \varnothing$ for every i, j.

01 **Show that $k \leq \ldots$**

Pair each subset with its complement. At most one member of each pair can be selected.

02 **Give a family attaining equality.**

All subsets containing 1: there are 2^{n-1} of them.

OPTIONAL CHALLENGE · SOLUTION

Complement pairs force the bound

For $n \geq 1$, a subset never equals its own complement.

Count the pairs

2^n subsets form 2^{n-1} pairs $\{A, U \setminus A\}$.

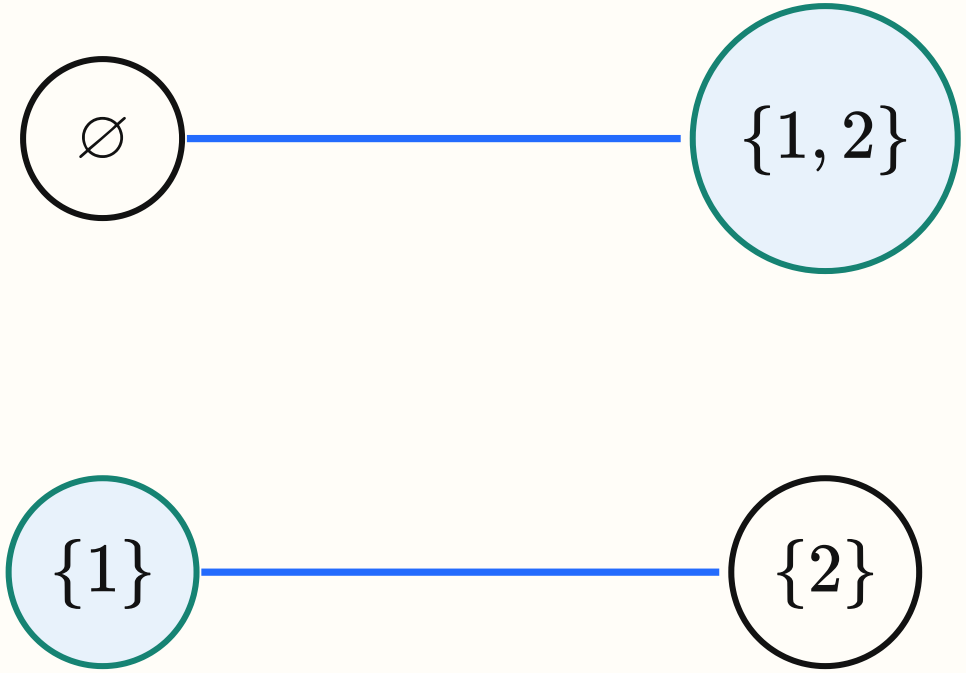
At most one per pair

$A \cap (U \setminus A) = \varnothing$, so $k \leq 2^{n-1}$.

Achieve equality

Select every subset containing 1; choose the other $n - 1$ elements freely.

Example: $U = \{1, 2\}$

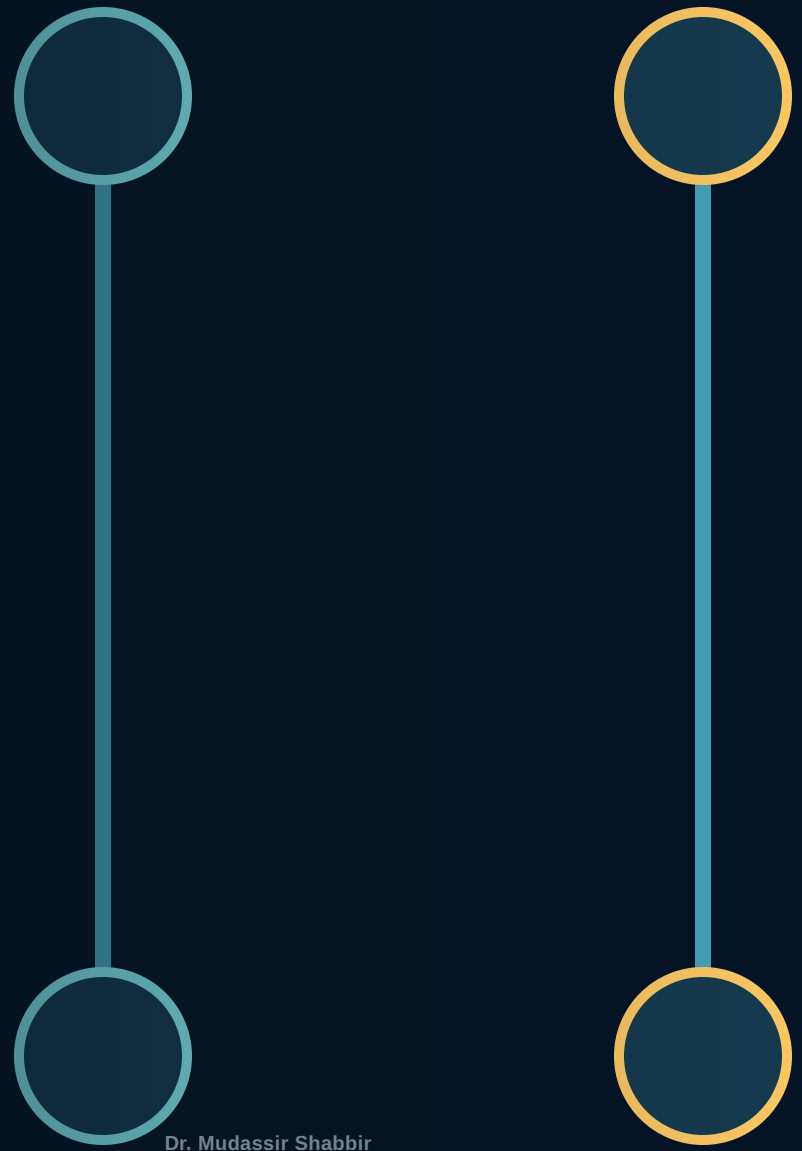


Lines pair complements, not order comparisons. Highlighted: all subsets containing 1.

TAKEAWAY

Order can leave pairs incomparable

Incomparability is different from equality: 8 and 12 are distinct maximal elements under divisibility.



LECTURE 17 · DISCRETE MATHEMATICS

One input. Exactly one output.

When does a relation deserve to be called a function?



MINDS BEHIND THE MATHEMATICS

Peter Dirichlet

1805–1859 · Germany

A tiny counting fact became a universal proof tool.

Original AI-illustrated abstract digital artwork

DIRICHLET · THE PERSON

Pigeons—or drawers?

In several languages, the pigeonhole principle is the drawer principle.

Objects

The things being placed

Boxes

The categories receiving them

The birds are optional. The collision is inevitable.

Original AI-illustrated abstract digital artwork

DIRICHLET · THE IDEA

Too many objects, not enough boxes

Place n + 1 objects into n boxes.

> some box has ≥ 2

Existence can be proved without finding the object.

Original AI-illustrated abstract digital artwork

Peter Gustav Lejeune Dirichlet

DIRICHLET · THE LEGACY

A simple idea with sharp consequences

The principle finds repetitions in strings, graphs, schedules, and number theory.

Move

Choose the right objects

Move

Choose the right boxes

The creativity lies in deciding what counts as a box.

Original AI-illustrated abstract digital artwork

Peter Gustav Lejeune Dirichlet

EXERCISE 8

A relation alone makes no output promise

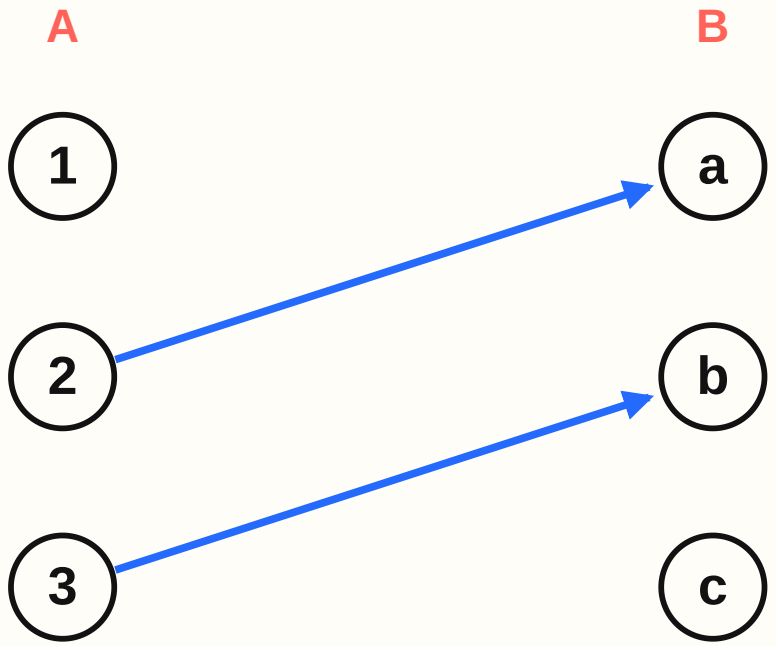
Let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$.
Inspect this relation from A to B.

01 **Does every input have an output?**

No: input 1 has no output.

02 **Is it still a relation? A function?**

It is a relation (a subset of the Cartesian product), but not a function on A.



01 · RECOGNIZING FUNCTIONS

Every input gets one arrow

The promise has two parts: existence and uniqueness.



Exactly one means at least one and at most one

A function $f : A \to B$ assigns exactly one element of B to each element of A.

Existence

For every $a \in A$, some $b \in B$ satisfies $f(a) = b$.

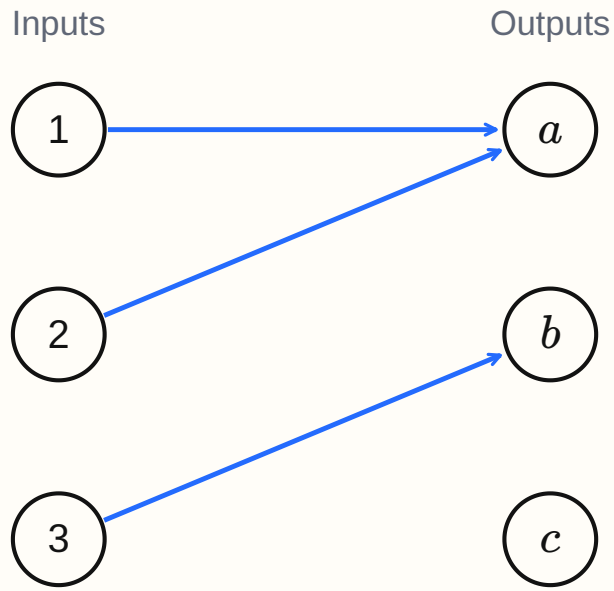
No input is left unmapped.

Uniqueness

$$f(a) = b \ \wedge \ f(a) = c$$

Then $b = c$: one input cannot split into different outputs.

Two inputs may share an output



Each input has exactly one arrow.

Different inputs are allowed to share the same output.

The quantifiers inspect each input

For a relation $R \subseteq A \times B$, the function condition is:

$$\forall a \in A \; \exists! \, b \in B : \; (a, b) \in R$$

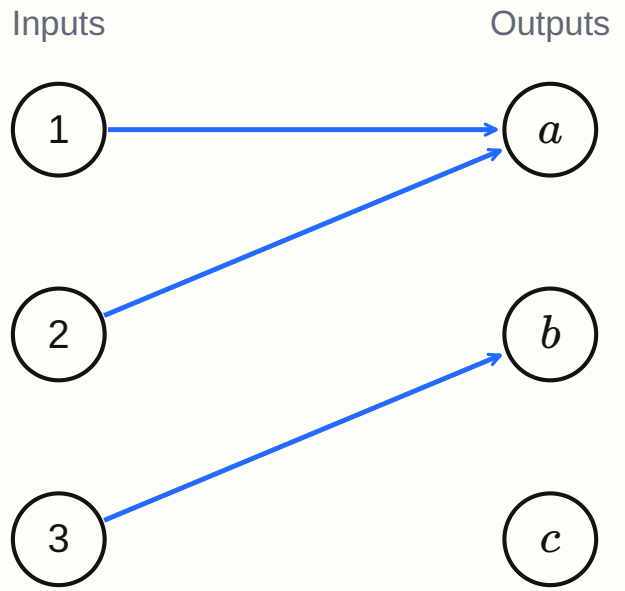
Existence

$$\forall a \in A \; \exists b \in B : \; (a, b) \in R$$

Uniqueness

$$(a, b) \in R \; \wedge \; (a, c) \in R \implies b = c$$

Inspect the arrows leaving each input



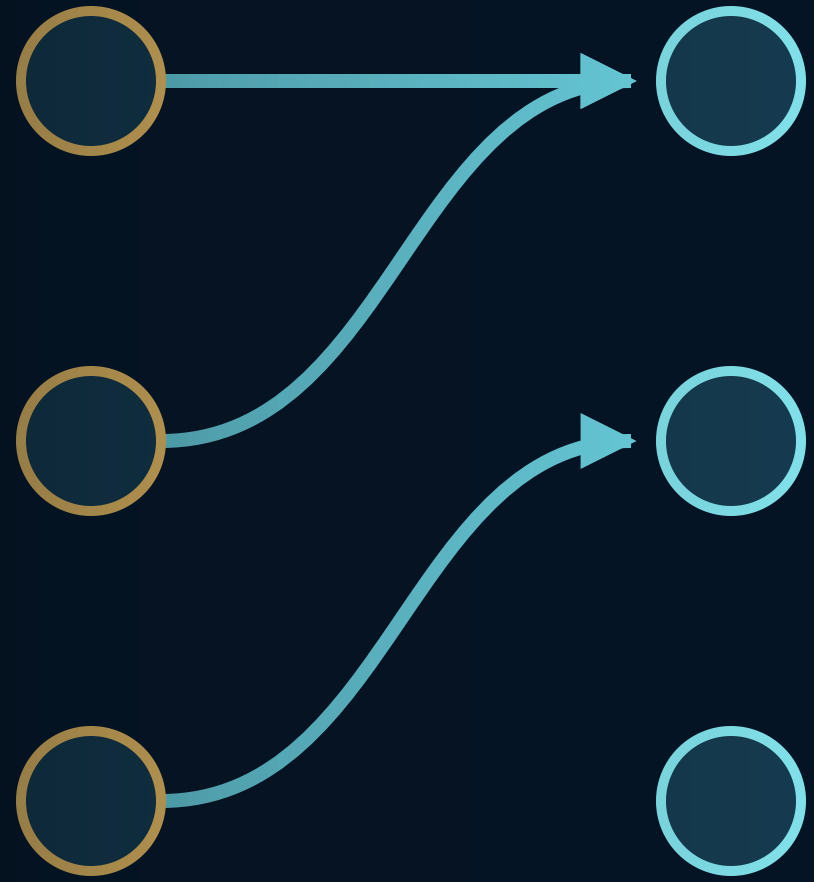
Each input has exactly one arrow.

Write $f(a) = b$ for the unique pair (a, b).

02 · FOLLOWING THE ARROWS

Allowed outputs versus actual outputs

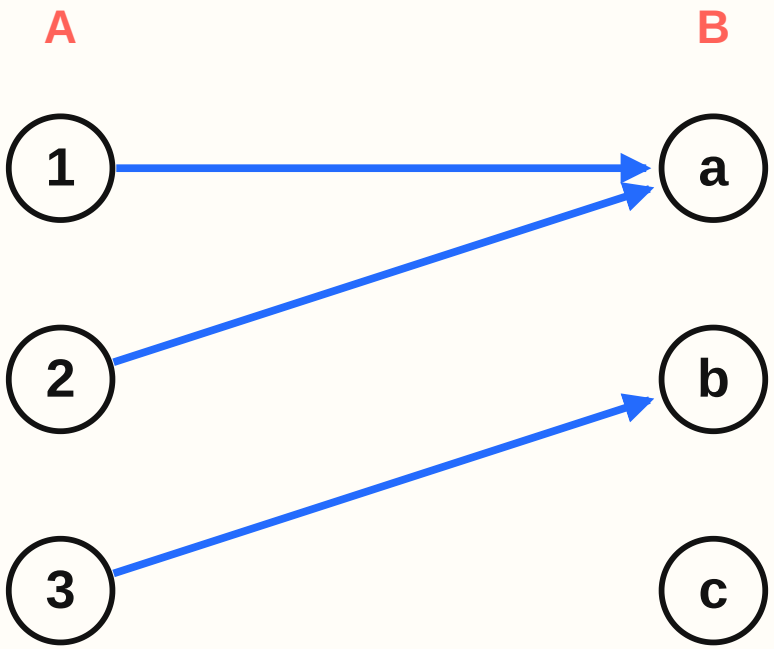
Domain, codomain, range, image, and preimage name different things.



Three sets; three different roles

For $f : A \to B$ shown here:

$A = \{1, 2, 3\}$
$B = \{a, b, c\}$
$f(A) = \{a, b\}$



Domain: A. Codomain: B. Range: the outputs actually reached.

EXERCISE 9

An image is forward; preimages are backward

Use the arrows to answer each question.

01 **What is the image of 2?**

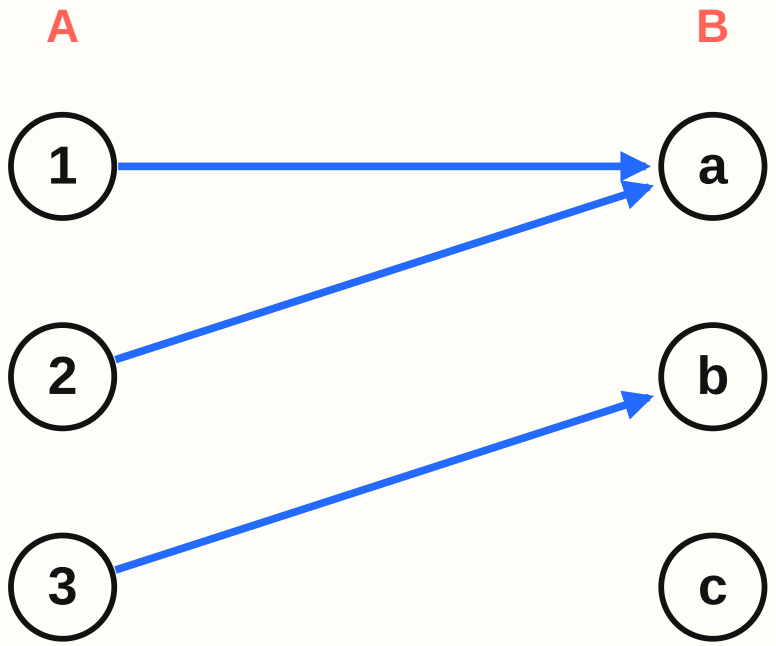
$f(2) = a$.

02 **Which inputs are preimages of a?**

1 and 2. Both map to a.

03 **Which inputs are preimages of c?**

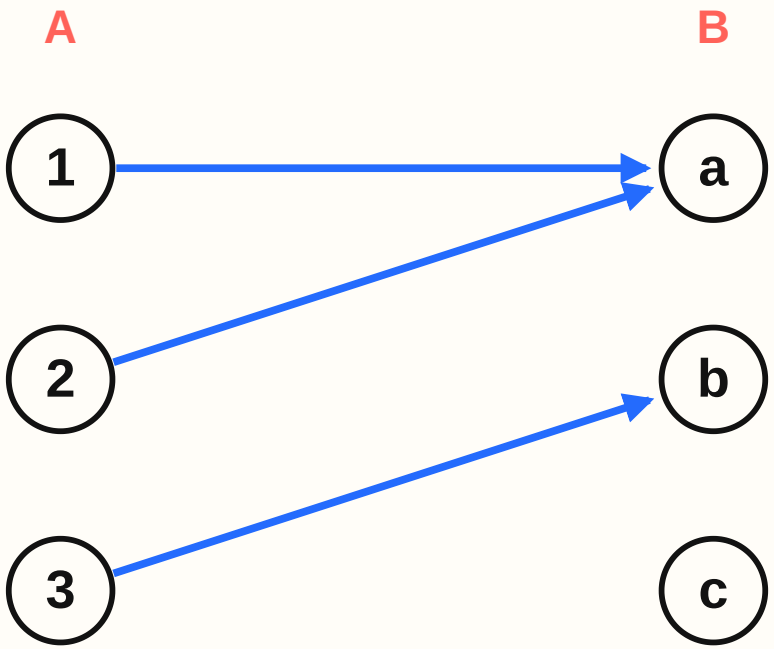
None: c is outside the range.



The image of a set collects its outputs

For $S \subseteq A$, collect its outputs:

$$f(S) = \{f(x) : x \in S\}$$
$$f(\{1, 2\}) = \{a\}$$
$$f(\{2, 3\}) = \{a, b\}$$



Repeated outputs appear only once in an image set.

A preimage set collects every matching input

For $T \subseteq B$, define:

$$f^{-1}(T) = \{x \in A : f(x) \in T\}$$

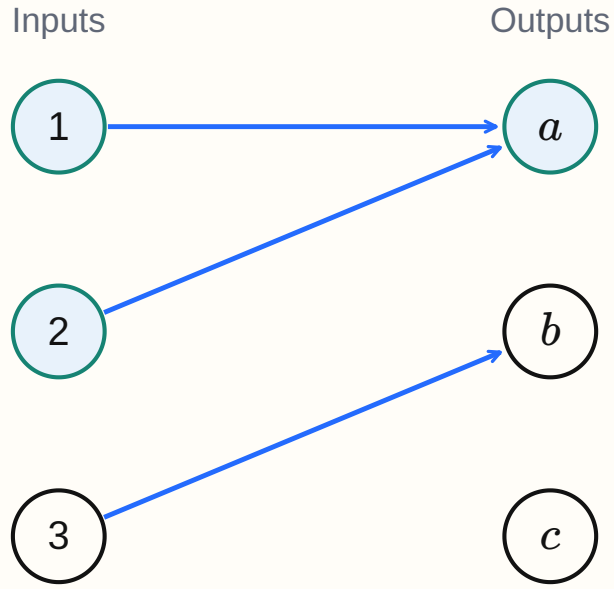
Start with targets

Choose a subset T of the codomain.

Search backward

Collect every input whose output belongs to T.

The preimage of $T = \{a\}$



$$f^{-1}(\{a\}) = \{1, 2\}.$$

This notation names a set; it does not assume an inverse function exists.

EXERCISE 10

Zero, one, or several preimages

Use the same function $f : A \to B$.

01 **Find $f^{-1}(\{a\})$.**

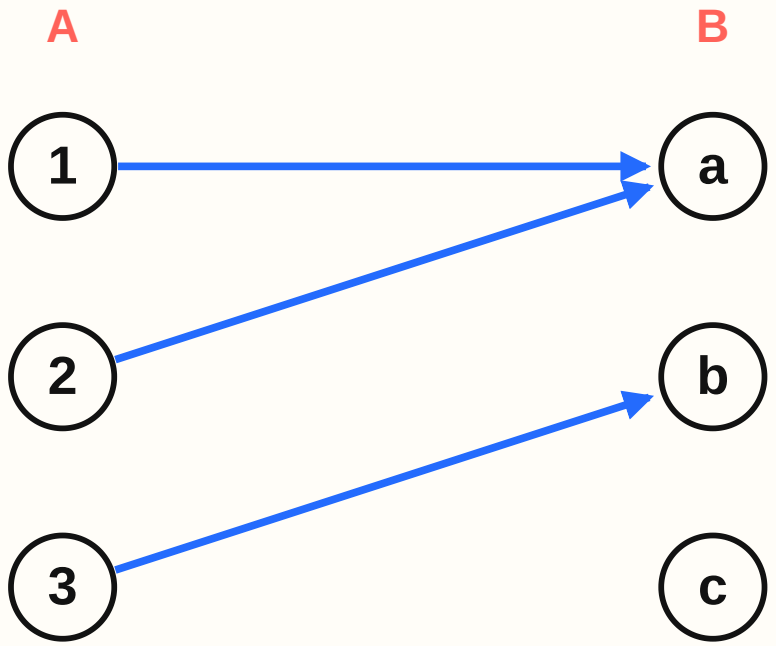
$\{1, 2\}$.

02 **Find $f^{-1}(\{b\})$.**

$\{3\}$.

03 **Find $f^{-1}(\{c\})$.**

$\varnothing$.



EXERCISE 11

Track a set through the arrows

Use $f(1) = a$, $f(2) = a$, and $f(3) = b$.

01 **Find $f(\{1, 3\})$.**

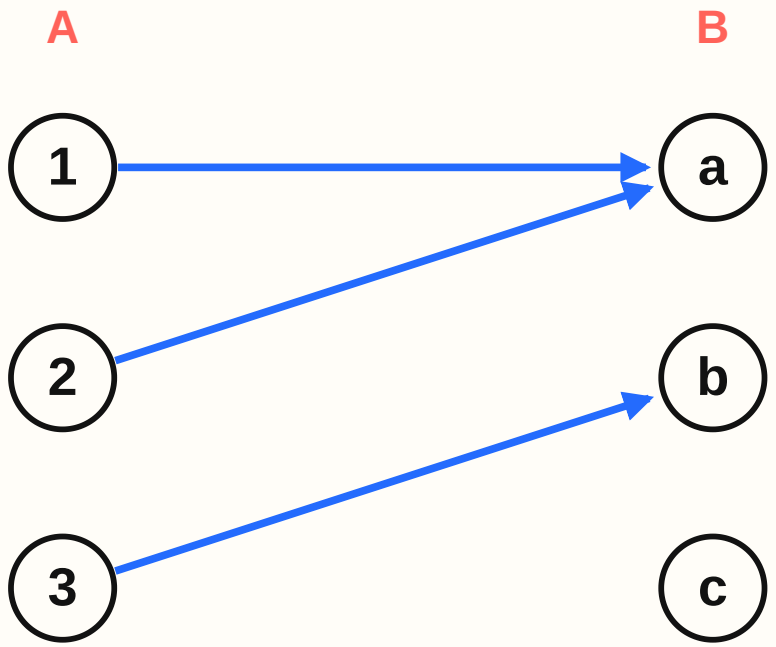
$\{a, b\}$.

02 **Find $f^{-1}(\{a, c\})$.**

$\{1, 2\}$; **the unused target c contributes no inputs.**

03 **Find $f^{-1}(f(\{1\}))$.**

$f^{-1}(\{a\}) = \{1, 2\}$. **It need not return only the starting input.**



The range of squaring contains only squares

For $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = x^2$:

Domain

$\mathbb{Z}$

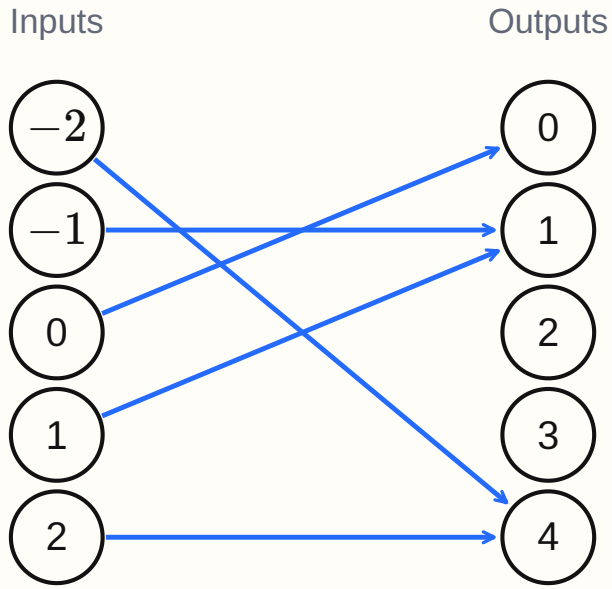
Codomain

$\mathbb{Z}$

Range

$f(\mathbb{Z}) = \{n^2 : n \in \mathbb{Z}\} = \{0, 1, 4, 9, \ldots\}$

A finite sample of integer squaring



2 and 3 are allowed targets, but neither is a square.

The range is not all integers, and not all nonnegative integers.

Changing the domain changes what is reached

Both rules square their inputs.

Integer inputs

$$f : \mathbb{Z} \to \mathbb{R}, \quad f(x) = x^2$$

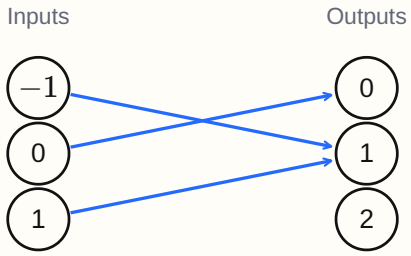
Range: $\{0, 1, 4, 9, \ldots\}$. No input maps to 2.

Real inputs

$$g : \mathbb{R} \to \mathbb{R}, \quad g(x) = x^2$$

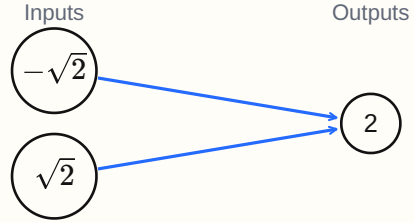
Range: $[0, \infty)$. Both $\sqrt{2}$ and $-\sqrt{2}$ map to 2.

Integer inputs cannot reach 2



A finite sample; no integer square equals 2.

Real inputs can reach 2



Both roots are real inputs.

EXERCISE 12

Check the sets before substituting

Does each declaration define a function? If not, repair it.

01 $f : \mathbb{R} \to \mathbb{R},\ f(x) = 1/x$

No: input 0 has no output. Use domain $\mathbb{R} \setminus \{0\}$.

---

02 $g : \mathbb{Z} \to \mathbb{Z},\ g(n) = n/2$

No: input 1 would produce $1/2 \notin \mathbb{Z}$. Use codomain $\mathbb{Q}$, or restrict the domain to even integers.

---

03 $h : \mathbb{R} \to \mathbb{R},\ h(x) = \sqrt{x}$

No over the reals: negative inputs have no real output. Use domain $[0, \infty)$.

A function includes its domain and codomain

Rosen treats two functions as equal only when their domains, codomains, and assignments all agree.

Same formula; different codomains

$$f : \mathbb{Z} \to \mathbb{Z}, \quad f(x) = x^2$$

A different declared function

$$g : \mathbb{Z} \to \mathbb{R}, \quad g(x) = x^2$$

The outputs agree at every integer, but the declared functions differ.

EXERCISE 13

A whole interval can share one image

Let $f : \mathbb{R} \to \mathbb{Z}$, $f(x) = \lfloor x \rfloor$.

01 **Find all preimages of 2.**

$f^{-1}(\{2\}) = [2, 3)$. **Include 2; exclude 3.**

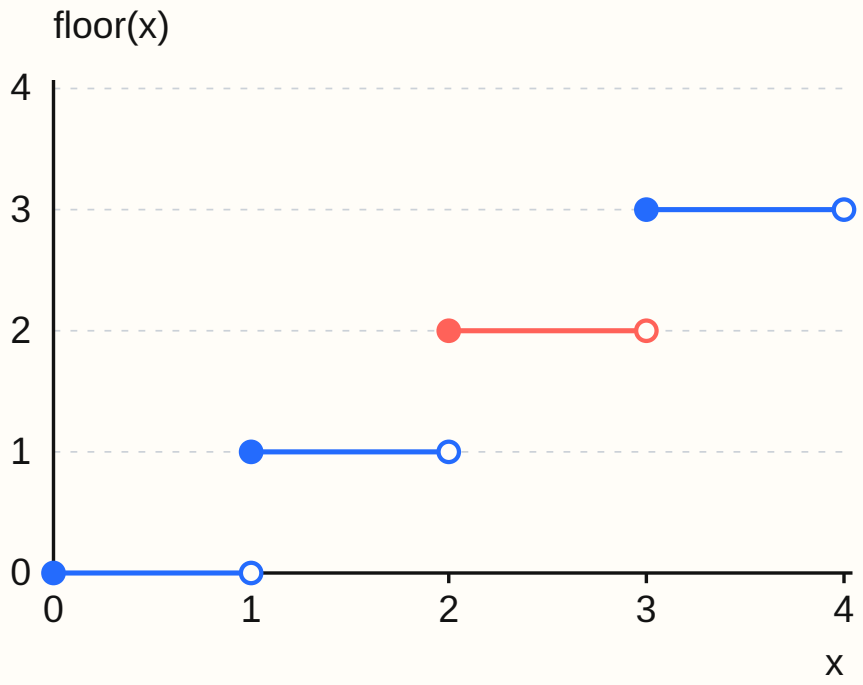
02 **Find the range of f.**

$\mathbb{Z}$: **every integer n is reached by input n.**

03 **Is this still a function?**

Yes. Each real input belongs to exactly one interval $[n, n + 1)$.

The graph of $y = \lfloor x \rfloor$



A filled endpoint is included; an open endpoint is excluded.

EXERCISE 14

Exit check: read the complete declaration

$f : \{-2, -1, 0, 1, 2\} \to \{0, 1, 2, 3, 4\}$ is defined by $f(x) = x^2$.

01 **Is it a function?**

Yes: each allowed input has exactly one square in the codomain.

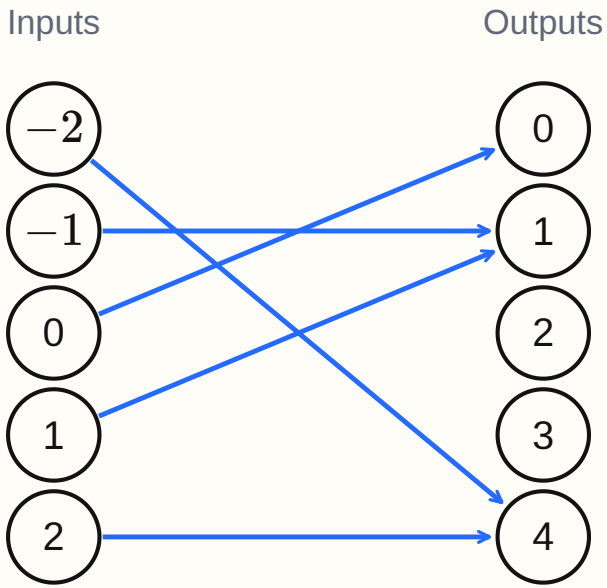
02 **What is its range?**

$\{0, 1, 4\}$.

03 **What are $f^{-1}(\{4\})$ and $f^{-1}(\{3\})$?**

$\{-2, 2\}$ and $\varnothing$, respectively.

The declared finite function

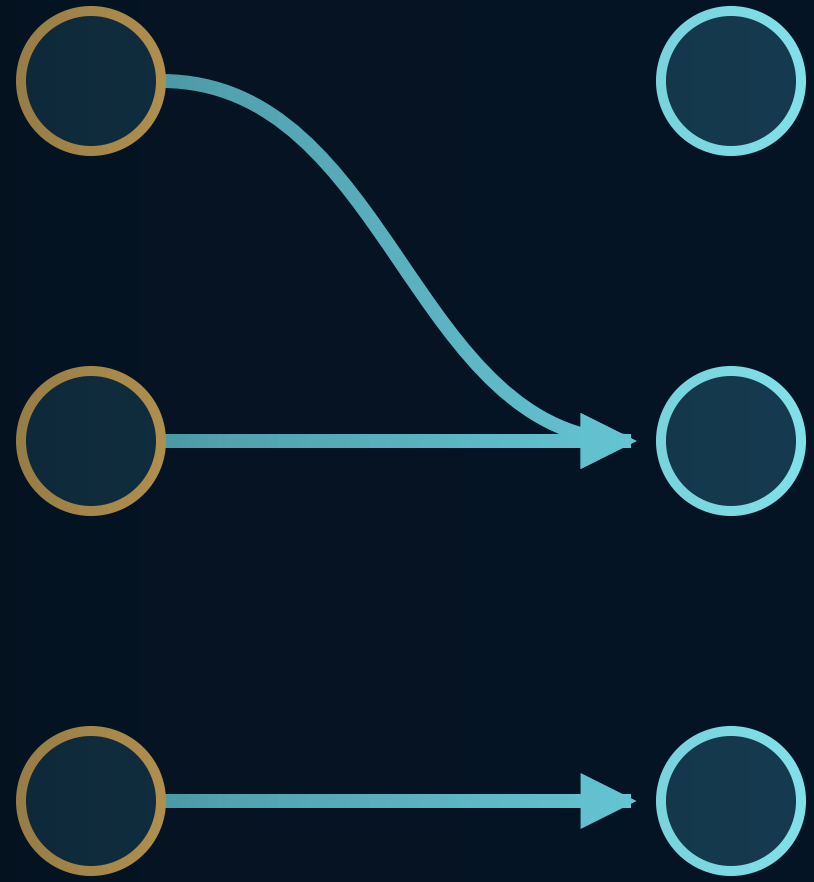


Follow the arrows to read the range and preimages.

TAKEAWAY

Every input keeps its promise

Exactly one output per input. Shared outputs and unused targets are allowed.

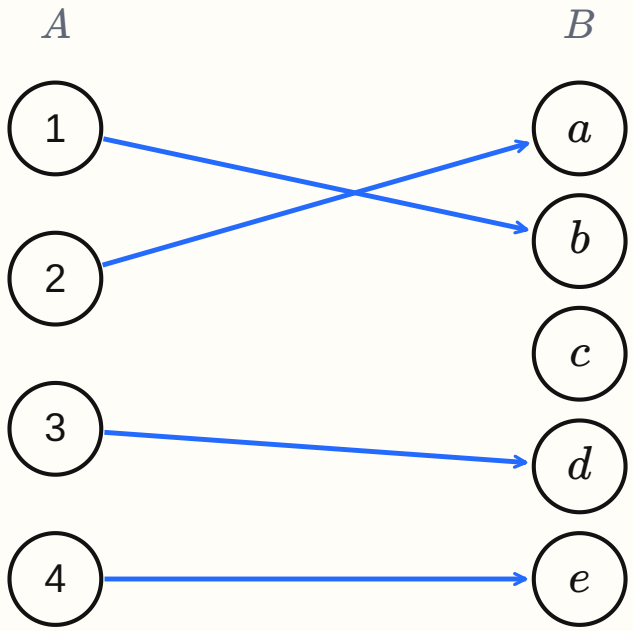


Injective functions have no collisions

A function $f : A \to B$ is injective when distinct inputs have distinct outputs.

$$f(x) = f(y) \implies x = y$$

Injective; one target remains unused



Target c is unused; each reached target has one preimage.

Every target has at most one preimage; unused targets are allowed.

EXERCISE 15

Squaring loses the sign

Let $f : \mathbb{Z} \to \mathbb{Z}_{\geq 0}$ be $f(x) = x^2$.

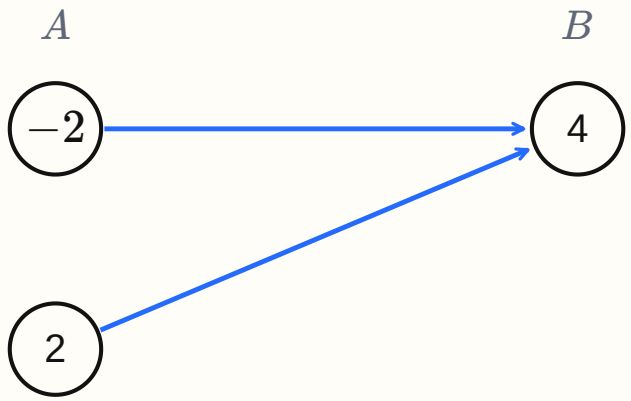
01 **Is f injective?**

No: $f(-2) = f(2) = 4$, while $-2 \neq 2$.

02 **Would nonnegative inputs remove this collision?**

Yes. On $\mathbb{Z}_{\geq 0}$, equality of squares forces equality of inputs.

A collision at 4



To prove injectivity, start with equal outputs

For $f : A \to B$, choose arbitrary $x, y \in A$.

Assume

$$f(x) = f(y)$$

Use the formula

Simplify until the inputs can be compared.

Conclude

$$x = y$$

To disprove injectivity, one pair of distinct inputs with equal outputs is enough.

The reciprocal function is injective

Let $X = \mathbb{R} \setminus \{0\}$ and $f : X \to X$, $f(x) = 1/x$.

Assume equal outputs

$$\frac{1}{x} = \frac{1}{y}$$

Multiply by the nonzero product

$$xy \cdot \frac{1}{x} = xy \cdot \frac{1}{y} \implies y = x$$

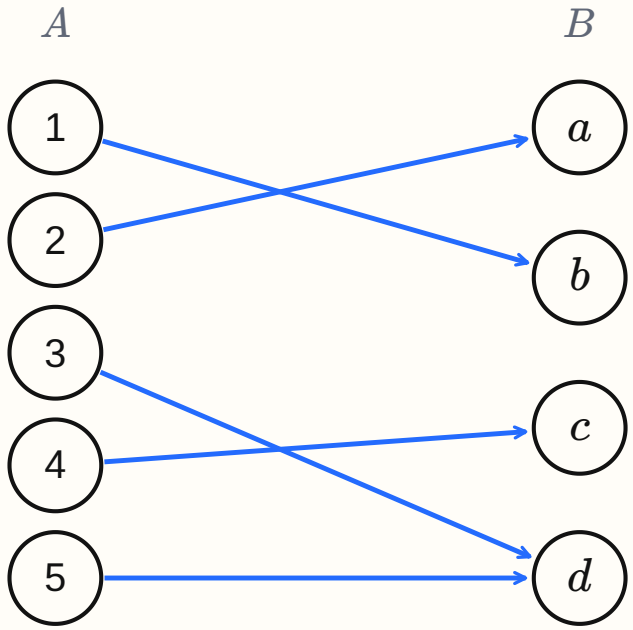
Both inputs are nonzero, so the multiplication is valid.

Surjective functions reach every target

A function $f : A \to B$ is surjective when $f(A) = B$.

$\forall b \in B \; \exists a \in A : \; f(a) = b$

Onto, with a shared output



Every target is reached; target d has two preimages.

Every codomain element has at least one preimage; collisions are allowed.

Surjectivity needs a preimage for an arbitrary target

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 1$.

Choose a target

Let $y \in \mathbb{R}$ be arbitrary.

Construct its preimage

$$x = (y - 1)/2 \in \mathbb{R}$$

Check the assignment

$$f(x) = 2((y - 1)/2) + 1 = y$$

The constructed input must belong to the declared domain.

A bijection gives exactly one preimage per target

A function is bijective when it is both injective and surjective.

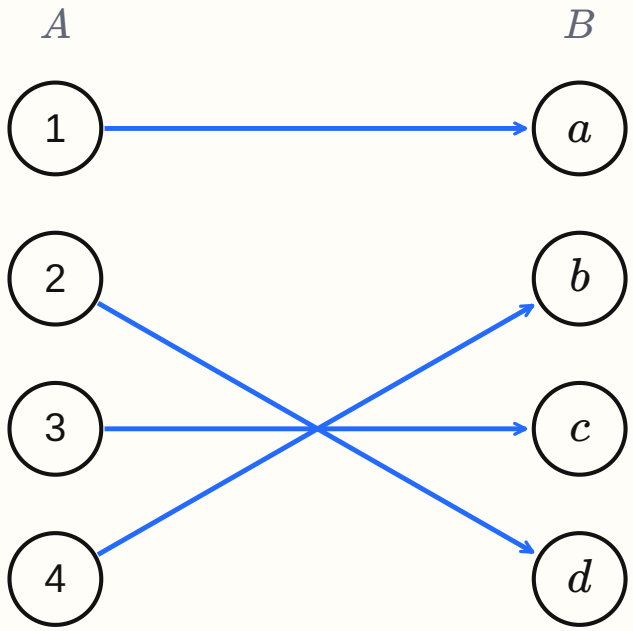
Injective

At most one preimage per target.

Surjective

At least one preimage per target.

A one-to-one correspondence



Every target has exactly one preimage.

Together: every target identifies one input.

EXERCISE 16

Equal finite sizes link the two properties

Suppose A and B are finite and $|A| = |B|$.

01 **If f is injective, can a target remain unused?**

No: $|A|$ distinct outputs already fill all $|B|$ targets.

02 **If f is surjective, can two inputs collide?**

No: a collision would leave too few distinct outputs to fill the codomain.

03 **If an injection has |A| = |range|, what about |B|?**

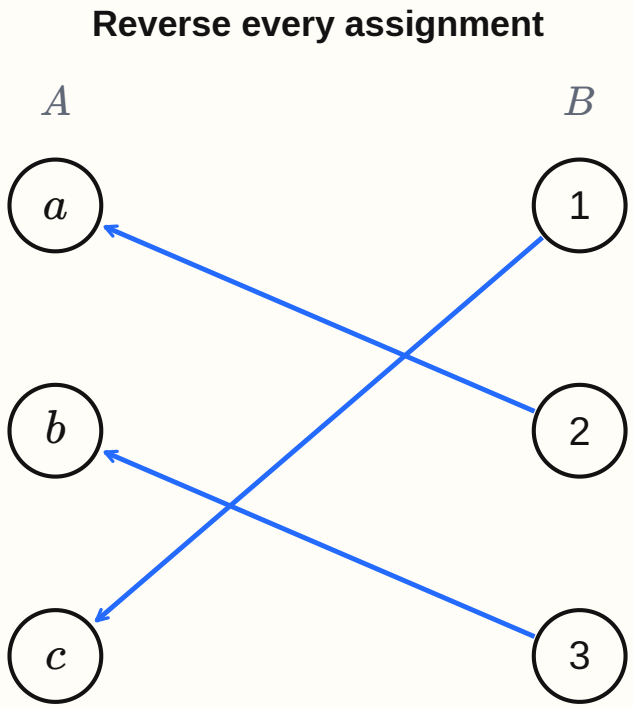
Only $|B| \geq |A|$ is forced; unused targets may exist.

For equal finite sets: injective ⇔ surjective ⇔ bijective.

An inverse reverses a bijection

If $f : A \to B$ is bijective, then $f^{-1} : B \to A$ satisfies:

$$f^{-1}(b) = a \iff f(a) = b$$



The displayed arrows represent the inverse.

A collision gives two reverse outputs; an unused target gives none.

EXERCISE 17

Find an inverse and check both directions

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 1$.

01 **Solve $y = 2x + 1$ for x.**

$f^{-1}(y) = (y - 1)/2$.

02 **Check $f^{-1}(f(x))$.**

$(2x + 1 - 1)/2 = x$.

03 **Check $f(f^{-1}(y))$.**

$2((y - 1)/2) + 1 = y$.

f^{-1} does not mean the reciprocal $1/f$.

Restricting both sets makes squaring reversible

Use $f : [0, \infty) \to [0, \infty)$, $f(x) = x^2$.

No sign collision

Nonnegative inputs with equal squares are equal.

No missed target

Every $y \geq 0$ has the input $\sqrt{y} \geq 0$.

Inverse

$$f^{-1}(y) = \sqrt{y}$$

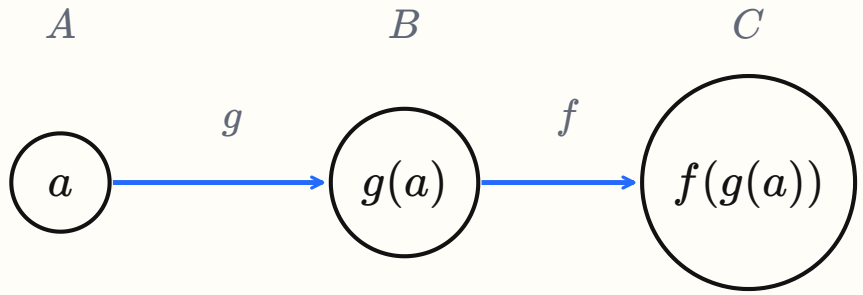
Changing the declared domain or codomain changes the function.

Composition follows the rightmost function first

Let $g : A \to B$ and $f : B \to C$.

$$(f \circ g)(a) = f(g(a))$$

First g, then f



Apply g first, then f: $A \to B \to C$.

The output of g must be a valid input to f.

EXERCISE 18

Trace a finite composition

$g : \{a, b, c\} \to \{a, b, c\}$ sends $a \mapsto b$, $b \mapsto c$, $c \mapsto a$. f sends $a \mapsto 3$, $b \mapsto 2$, $c \mapsto 1$.

01 **Find $(f \circ g)(a), (f \circ g)(b), (f \circ g)(c)$.**

2, 1, 3 respectively.

---

02 **Is $g \circ f$ defined here?**

No: outputs 1, 2, 3 are not in the domain of g.

An inverse cancels on the appropriate set

For a bijection $f : A \to B$:

Start in A

$$f^{-1} \circ f = \mathrm{id}_A$$

Start in B

$$f \circ f^{-1} = \mathrm{id}_B$$

The identity function returns each element unchanged.

EXERCISE 19

Exit check: classify the full declaration

$f : \mathbb{Z} \to \mathbb{Z}$ is $f(n) = n + 1$.

01 **Is f a bijection?**

Yes. Equal outputs give equal inputs, and target m has preimage m-1.

02 **Give its inverse.**

$f^{-1}(m) = m - 1$.

03 **What changes for $f : \mathbb{Z}_{\geq 0} \to \mathbb{Z}_{\geq 0}$?**

It stays injective but misses 0, so it is not onto.

TAKEAWAY

Reversibility needs exactly one preimage

Injective excludes collisions. Surjective excludes missed targets.
A bijection lets every arrow be reversed.